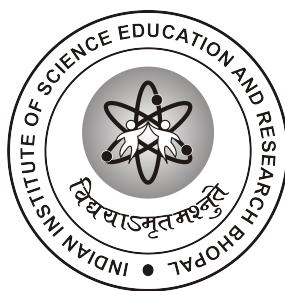


MTH 518/618  
Commutative Algebra

# Relationship between Artinian and Noetherian Rings

By Al Amin Miah  
Roll Number: 230412



Department of Mathematics, IISERB

November 4, 2024

## Introduction

The following is a compilation of the notes that I prepared for my presentation required for the completion of the course MTH 518/618 (Commutative Algebra) under the supervision of Dr. Sanjay Kumar Singh. I have primarily used [?] as my *prime* reference and used definitions and results from it to build almost all machinery required to prove our main theorem, “**A ring is Artinian  $\iff$  it is Noetherian and is of dimension zero**”. I have also included comments in *italics* before each result to give a brief idea on the requirement of such a result and to motivate further reading.

*Note that all rings are taken to be commutative and with unity and ring homomorphisms always maps unity to unity. Dimension of a ring means the Krull dimension of the ring. Consider the ring to be  $A$  wherever not mentioned and  $M$  as an  $A$ -module, unless otherwise stated.*

*Understanding how prime ideals interact with intersections of ideals is crucial in many structural results. Here, we establish that if a prime ideal contains an intersection of finitely many ideals, then it must contain at least one of them.*

**Proposition 1.** *Let  $\mathfrak{a}_1, \mathfrak{a}_2 \dots \mathfrak{a}_n$  be ideals and  $\mathfrak{p}$  be a prime ideal containing  $\bigcap_{i=1}^n \mathfrak{a}_i$ . Then  $\mathfrak{p} \supseteq \mathfrak{a}_i$  for some  $i$ .*

*Proof.* Suppose  $\mathfrak{p} \not\supseteq \mathfrak{a}_i \forall i$ . Then  $\exists x_i \in \mathfrak{a}_i \setminus \mathfrak{p} (1 \leq i \leq n)$ . Thus  $\prod_{i=1}^n x_i \in \prod_{i=1}^n \mathfrak{a}_i \subseteq \bigcap_{i=1}^n \mathfrak{a}_i$ ; but  $\prod_{i=1}^n x_i \notin \mathfrak{p}$  (since  $\mathfrak{p}$  is prime). Hence  $\bigcap_{i=1}^n \mathfrak{a}_i \not\subseteq \mathfrak{p}$ , a contradiction.  $\square$

*Modules often have complex submodule structures, and one way to measure their complexity is through chains of submodules. A composition series provides a maximal such chain where each successive quotient is simple. Studying these series allows us to classify modules based on their ‘length’, which will be critical in distinguishing Noetherian and Artinian behavior.*

**Definition 1** (Chain). A chain of submodules of length  $n$  of an  $A$ -module  $M$  is a sequence of submodules  $\{M_i\}_{i=0}^n$  of it such that

$$\{0\} = M_0 \subsetneq M_1 \subsetneq M_2 \subsetneq \dots \subsetneq M_{n-1} \subsetneq M_n = M \quad (1)$$

**Definition 2** (Composition Series). A composition series is a maximal chain of submodules, i.e., a chain in which no additional submodules can be inserted or, simply put,  $M_{i+1}/M_i$  is simple  $\forall i \in \{0, 1, \dots, n-1\}$  in (1).

*When a module admits a composition series, we show that all such series must have the same length. This suggests that ‘length’ is an intrinsic property of the module rather than an arbitrary choice of submodules. Later, we will see how this connects to Noetherian and Artinian conditions, which both impose constraints on submodule chains.*

**Proposition 2.** *If  $M$  has a composition series of length  $n$ , then every other composition series of  $M$  is also of length  $n$  and every chain in  $M$  can be extended to a composition series.*

*Proof.* Let  $l(\bullet) :=$  the least length of a composition series in  $\bullet$ .  
Observe that  $l(M) = \infty \iff M$  has no composition series.

Claim 1: If  $N \subsetneq M$  then  $l(N) < l(M)$

Let  $(M_i)$  be a composition series of length  $l(M)$ . Let  $N_i = M_i \cap N$ , then  $N_i$  is a submodule of  $N$ . Also we have

$$\begin{aligned}
& N_{i+1}/N_i \xrightarrow{\phi} M_{i+1}/M_i \\
& \Rightarrow \phi(N_{i+1}/N_i) \subseteq M_{i+1}/M_i \\
& \Rightarrow \phi(N_{i+1}/N_i) = 0 \quad (\text{Since } M_{i+1}/M_i \text{ is simple}) \\
& \Rightarrow \phi(N_{i+1}/N_i) = M_{i+1}/M_i \\
& \Rightarrow N_{i+1}/N_i = 0 \text{ or } N_{i+1}/N_i \cong M_{i+1}/M_i \\
& \Rightarrow N_{i+1} = N_i \text{ or } N_{i+1}/N_i \text{ is simple}
\end{aligned}$$

Hence by removing repeated terms we get a composition series in  $N$  such that  $l(N) \leq l(M)$

Now, if  $l(N) = l(M)$  then  $N_{i+1} \neq N_i \forall i$  so we get,

$$\begin{aligned}
& N_{i+1}/N_i \cong M_{i+1}/M_i \quad \forall i \\
& \Rightarrow N_1/N_0 \cong M_1/M_0 = 0 \\
& \Rightarrow N_1 \cong M_1 \\
& \Rightarrow N_1 = M_1
\end{aligned}$$

Thus

$$\begin{aligned}
& N_2/N_1 \cong M_2/M_1 \\
& \Rightarrow N_2 = M_2
\end{aligned}$$

Similarly,

$$\begin{aligned}
& N_3 = M_3 \\
& \vdots \\
& N_n = M_n \\
& \Rightarrow N = M
\end{aligned}$$

which is a contradiction. Hence the claim.

Claim 2: Any chain in  $M$  has length  $\leq l(M)$

Let us consider a chain of length  $k$  in  $M$

$$0 = M_0 \subsetneq M_1 \subsetneq M_2 \subsetneq \dots \subsetneq M_{k-1} \subsetneq M_k = M$$

Then,

$$\begin{aligned}
0 < l(M_1) < l(M_2) < \dots < l(M_{k-1}) < l(M) && \text{(By Claim 1)} \\
\Rightarrow l(M) \geq k && \text{(Since } l(M) \text{ is strictly greater than } k \text{ positive intergers)}
\end{aligned}$$

Hence the claim.

Now, consider a composition series of length  $k$  then  $k \leq l(M)$  (By Claim 2) and by definition  $k \geq l(M)$ . Thus we have  $k = l(M)$  i.e., any composition series in  $M$  has length  $l(M)$ .

To conclude, take any chain in  $M$  if its length is  $l(M)$  then it is a composition series if not then it is not a composition series hence not maximal thus new terms can be inserted in between wherever required until its length is  $l(M)$ .  $\square$

*Artinian and Noetherian modules are characterized by their ascending (acc) and descending chain conditions (dcc), respectively. This result establishes a crucial link: a module has a composition series if and only if it satisfies both acc and dcc. This insight will later help us prove that Artinian rings have well-controlled ideal structures.*

**Proposition 3.**  *$M$  has a composition series iff  $M$  satisfies both acc and dcc*

*Proof.* ( $\Rightarrow$ ) Consider a chain of submodules of  $M$

$$0 = M_0 \subseteq M_1 \subseteq M_2 \dots$$

If the above chain does not become stationary after finitely many terms then we can find a sub-chain  $\{M_{i_n}\}$  of the above chain such that

$$0 = M_0 \subsetneq M_{i_1} \subsetneq M_{i_2} \subsetneq \dots$$

But by *Claim 2* of Proposition 2 the length of any chain in  $M$  is bounded above by  $l(M) < \infty$  (since  $M$  has a composition series) contradicting the existence of such a chain with strict inclusions. Hence  $M$  satisfies acc. Similarly,  $M$  satisfies dcc as well.

( $\Leftarrow$ ) Consider  $\Sigma :=$  The set of all submodules contained in  $M$ . Then every subset of  $\Sigma$  has a maximal element (By acc). Thus the subset of  $\Sigma$  consisting of submodules which are strictly contained in  $M$  has a maximal element i.e.,  $\exists$  a maximal submodule strictly contained in  $M = M_n$  say  $M_{n-1}$ . We repeat the same process to get a decreasing maximal chain

$$M_n \supsetneq M_{n-1} \supsetneq M_{n-2} \dots$$

but  $M$  satisfies dcc. Hence the above chain reduces to

$$M_n \supsetneq M_{n-1} \supsetneq M_{n-2} \dots \supsetneq M_1 \supsetneq M_0 = 0$$

which is a composition series in  $M$ . □

*To build intuition, we first consider vector spaces, where the notions of acc and dcc become much more tangible. This proposition shows that for vector spaces, these conditions are equivalent to finite dimensionality. This analogy foreshadows the fundamental theorem that an Artinian ring behaves ‘finite-dimensionally’ in a certain sense.*

**Proposition 4.** *For a  $k$ -vector space  $V$  the following are equivalent:*

- (i)  $V$  is finite dimensional
- (ii)  $V$  has a composition series i.e.,  $V$  is of finite length
- (iii)  $V$  satisfies acc
- (iv)  $V$  satisfies dcc

*Proof.* ((i)  $\Rightarrow$  (ii)) For a chain of subspaces

$$0 = W_0 \subsetneq W_1 \subsetneq W_2 \subsetneq \dots$$

we have

$$0 < \dim(W_1) < \dim(W_2) < \dots$$

but  $\dim(V) < \infty \Rightarrow \exists k$  such that  $\dim(V) = \dim(W_k)$  and the chain reduces to

$$0 = W_0 \subsetneq W_1 \subsetneq W_2 \subsetneq \dots \subsetneq W_k = V$$

((ii)  $\Rightarrow$  (iii)) and ((iii)  $\Rightarrow$  (iv)) follows from Proposition 3.

((iii)  $\Rightarrow$  (i)) Suppose  $V$  is not of finite dimension. Consider  $\beta = \{x_1, x_2, x_3, \dots\}$  be a set of infinitely many linearly independent vectors in  $V$ . Then the chain

$$(x_1) \subsetneq (x_1, x_2) \subsetneq (x_1, x_2, x_3) \subsetneq \dots$$

never becomes stationary contradicting (iii) Hence  $V$  must be of finite dimension.  
 ((iv)  $\Rightarrow$  (i)) Suppose  $V$  is not of finite dimension and  $\beta$  is as above. Consider the chain

$$(x_{n+1}, x_{n+2}, \dots) \supsetneq x_{n+2}, x_{n+3}, \dots) \supsetneq (x_{n+3}, x_{n+4}, \dots) \supsetneq \dots$$

which never becomes stationary, a contradiction. Thus  $V$  is of finite dimension.  $\square$

*The following corollary provides a useful criterion for identifying Artinian rings. If the zero ideal factors as a product of maximal ideals, the ring satisfies both chain conditions, leading to the equivalence of Noetherian and Artinian properties. This insight will be crucial in our proof of the main theorem.*

**Corollary 4.1.** *Let  $A$  be a ring in which the zero ideal is a product  $\mathfrak{m}_1\mathfrak{m}_2\mathfrak{m}_3\dots\mathfrak{m}_n$  of not necessarily distinct maximal ideals. Then  $A$  is Noetherian iff  $A$  is Artinian.*

*Proof.* Observe that

$$0 \longrightarrow \mathfrak{m}_1 \longrightarrow A \longrightarrow A/\mathfrak{m}_1 \longrightarrow 0$$

is a short exact sequence and thus  $A$  is Noetherian (Artinian)  $A$ -module iff  $\mathfrak{m}_1$  and  $A/\mathfrak{m}_1$  are Noetherian (Artinian)  $A$ -module

Suppose  $A$  is Artinian. Then  $A/\mathfrak{m}_1$  is a  $A/\mathfrak{m}_1$ -vector space and  $A/\mathfrak{m}_1$  is Artinian so  $A/\mathfrak{m}_1$  is Artinian as  $A/\mathfrak{m}_1$ -module

$\Rightarrow A/\mathfrak{m}_1$  is a  $A/\mathfrak{m}_1$ -vector space and it satisfies dcc

$\Rightarrow A/\mathfrak{m}_1$  is a  $A/\mathfrak{m}_1$ -vector space and it satisfies acc (By Proposition 4)

$\Rightarrow A/\mathfrak{m}_1$  is a Noetherian  $A/\mathfrak{m}_1$ -module

$\Rightarrow A/\mathfrak{m}_1$  is a Noetherian  $A$ -module

Next, we have the following short exact sequence

$$0 \longrightarrow \mathfrak{m}_1\mathfrak{m}_2 \longrightarrow \mathfrak{m}_1 \longrightarrow \mathfrak{m}_1/\mathfrak{m}_1\mathfrak{m}_2 \longrightarrow 0$$

and as  $\mathfrak{m}_1$  is an Artinian  $A$ -module, thus we have  $\mathfrak{m}_1\mathfrak{m}_2$  and  $\mathfrak{m}_1/\mathfrak{m}_1\mathfrak{m}_2$  to be Artinian  $A$ -module as well

$\Rightarrow \mathfrak{m}_1/\mathfrak{m}_1\mathfrak{m}_2$  is an Artinian  $A/\mathfrak{m}_2$ -module

$\Rightarrow \mathfrak{m}_1/\mathfrak{m}_1\mathfrak{m}_2$  is an  $A/\mathfrak{m}_2$ -vector space that satisfies dcc

$\Rightarrow \mathfrak{m}_1/\mathfrak{m}_1\mathfrak{m}_2$  satisfies acc as well (By Proposition 4)

$\Rightarrow \mathfrak{m}_1/\mathfrak{m}_1\mathfrak{m}_2$  is a Noetherian  $A/\mathfrak{m}_2$ -module

$\Rightarrow \mathfrak{m}_1/\mathfrak{m}_1\mathfrak{m}_2$  is a Noethrian  $A$ -module.

Thus we just need to show that  $\mathfrak{m}_1/\mathfrak{m}_1\mathfrak{m}_2$  is a Noetherian  $A$ -module. However, proceeding

like above we will reach a stage where it will be enough to prove that  $\mathfrak{m}_1\mathfrak{m}_2\ldots\mathfrak{m}_{n-1}$  is Noetherian  $A$ -module provided  $\mathfrak{m}_1\mathfrak{m}_2\ldots\mathfrak{m}_{n-1}$  is Artinian  $A$ -module.

Thus we consider the short exact sequence

$$0 \longrightarrow \mathfrak{m}_1\mathfrak{m}_2\ldots\mathfrak{m}_n \longrightarrow \mathfrak{m}_1\mathfrak{m}_2\ldots\mathfrak{m}_{n-1} \longrightarrow \mathfrak{m}_1\mathfrak{m}_2\ldots\mathfrak{m}_{n-1}/\mathfrak{m}_1\mathfrak{m}_2\ldots\mathfrak{m}_n \longrightarrow 0$$

but  $\mathfrak{m}_1\mathfrak{m}_2\ldots\mathfrak{m}_n = 0 \Rightarrow \mathfrak{m}_1\mathfrak{m}_2\ldots\mathfrak{m}_{n-1}/\mathfrak{m}_1\mathfrak{m}_2\ldots\mathfrak{m}_n \cong \mathfrak{m}_1\mathfrak{m}_2\ldots\mathfrak{m}_{n-1}$  is a Noetherian  $A$ -module. Similarly, if  $A$  is Noetherian then using the same process as above we will arrive at the conclusion that  $A$  is Artinian as well.  $\square$

*Remark.* In the above proof we have used the following fact that for an  $A$ -module  $M$  and  $\mathfrak{a}$ , an ideal of  $A$ ,  $M/\mathfrak{a}M$  is both an  $A$  and an  $A/\mathfrak{a}$ -module and the following holds true:

1. The submodules of  $M/\mathfrak{a}M$  as an  $A$ -module are the same as those of it as an  $A/\mathfrak{a}$ -module
2.  $M/\mathfrak{a}M$  is Noetherian (Artinian) as an  $A$ -module iff it is Noetherian (Artinian) as an  $A/\mathfrak{a}$ -module

*One striking feature of Artinian rings is that prime ideals cannot be ‘deep’—they must already be maximal. This follows from the descending chain condition (dcc) and significantly simplifies the structure of Artinian rings, ensuring that their Krull dimension is always 0.*

**Lemma 5.** *In an Artinian ring every prime ideal is maximal*

*Proof.* Let  $\mathfrak{p}$  be a prime ideal of a ring  $A$ . Then  $B := A/\mathfrak{p}$  is an Artinian integral domain. Let  $0 \neq x \in B$ . By dcc  $(x^n) = (x^{n+1})$  for some  $n$ , hence  $x^n = x^{n+1}y$  i.e.,  $x^n(xy - 1) = 0$  for some  $y \in B$ . Since  $B$  is an integral domain and  $x \neq 0$  we must have  $xy = 1$  i.e.,  $A/\mathfrak{p}$  is a field and thus  $\mathfrak{p}$  is maximal.  $\square$

*Since all prime ideals in an Artinian ring are maximal, the nilradical (intersection of all prime ideals) coincides with the Jacobson radical (intersection of all maximal ideals). This observation further reinforces the ‘finiteness’ properties of Artinian rings.*

**Corollary 5.1.** *In an Artinian ring the nilradical  $\mathfrak{N}$  is equal to the Jacobson radical  $\mathfrak{J}$*

*Unlike general Noetherian rings, where the number of maximal ideals can be infinite, Artinian rings have a strict constraint: they can only have finitely many maximal ideals.*



**Lemma 6.** *An Artinian ring has only a finite number of maximal ideals*

*Proof.* Consider the set  $\Sigma := \{\mathfrak{m}_1 \cap \mathfrak{m}_2 \cap \cdots \cap \mathfrak{m}_r \mid \mathfrak{m}_i \text{ is a maximal ideal, } r \geq 0\}$ . This set has a minimal element say  $\mathfrak{m}_1 \cap \mathfrak{m}_2 \cap \cdots \cap \mathfrak{m}_n$ . Then for any maximal ideal  $\mathfrak{m}$  we have  $\mathfrak{m} \cap \mathfrak{m}_1 \cap \mathfrak{m}_2 \cap \cdots \cap \mathfrak{m}_n = \mathfrak{m}_1 \cap \mathfrak{m}_2 \cap \cdots \cap \mathfrak{m}_n$ . By Proposition 1,  $\mathfrak{m} \supseteq \mathfrak{m}_i$  for some  $i \in \{1, 2, \dots, n\}$  but  $\mathfrak{m}_i$  is maximal and hence  $\mathfrak{m} = \mathfrak{m}_i$ .  $\square$

*In an Artinian ring, the nilradical is not just an ideal that contains nilpotent elements—it itself must be nilpotent. This result is an important step toward proving that the ring has a well-controlled structure and will help establish that Artinian rings are Noetherian.*

**Lemma 7.** *In an Artinian ring the nilradical  $\mathfrak{R}$  is nilpotent*

*Proof.* Consider the decreasing chain

$$\mathfrak{R} \supseteq \mathfrak{R}^2 \supseteq \mathfrak{R}^3 \supseteq \dots$$

As  $A$  is Artinian  $\exists k > 0$  such that  $\mathfrak{a} := \mathfrak{R}^k = \mathfrak{R}^{k+1} = \mathfrak{R}^{k+2} = \dots$

Let us assume that  $\mathfrak{a} \neq 0$  and consider  $\Sigma' := \{\mathfrak{b} \text{ an ideal of } A \mid \mathfrak{b}\mathfrak{a} \neq 0\}$ . Clearly  $\mathfrak{R} \in \Sigma'$  and thus  $\Sigma' \neq \emptyset$ . Now, as  $A$  is Artinian  $\Sigma'$  has a minimal element say  $\mathfrak{c}$ . Then  $\mathfrak{c}\mathfrak{a} \neq 0$  and thus  $\exists x \in \mathfrak{c}$  such that  $x\mathfrak{a} \neq 0$

$$\begin{aligned} &\Rightarrow xa \neq 0 && \forall a \in \mathfrak{a} \\ \Rightarrow bxa = xba = xa' \neq 0 && \forall b \in A, a \in \mathfrak{a} \\ \Rightarrow (x)\mathfrak{a} \neq 0 \\ \Rightarrow (x) \in \Sigma' \\ \Rightarrow (x) = \mathfrak{c} && \text{(By minimality of } \mathfrak{c}) \end{aligned}$$

Again observe that

$$x\mathfrak{a}\mathfrak{a} = x\mathfrak{a}^2 = x\mathfrak{a} \neq 0$$

Hence

$$\begin{aligned} x\mathfrak{a} &= (x) && \text{(By minimality of } \mathfrak{c} = (x)) \\ \Rightarrow x &= xy \text{ for some } y \in \mathfrak{a} \\ \Rightarrow x &= xy = xy^2 = xy^3 = \cdots = xy^n = \dots \end{aligned}$$

But

$$\begin{aligned}
y &\in \mathfrak{a} = \mathfrak{R}^k \subseteq \mathfrak{R} \\
&\Rightarrow y^n = 0 \text{ for some } n > 0 \\
&\Rightarrow x = xy^n = 0 \\
&\Rightarrow x\mathfrak{a} = 0, \text{ a contradiction}
\end{aligned}$$

Hence we must have  $\mathfrak{a} = 0$  □

*The radical of an ideal consists of elements whose powers eventually land in the ideal, but in general, this doesn't mean the ideal itself must contain large powers of its radical. However, in a Noetherian ring, this is exactly what happens: some power of the radical is always forced inside the ideal. This result reflects the ascending chain condition in Noetherian rings and is particularly useful when studying nilpotency. In the special case where the ideal is the zero ideal, this directly implies that the nilradical is nilpotent, bridging the gap between Noetherian and Artinian properties.*

**Lemma 8.** *In a Noetherian ring every ideal contains a power of its radical*

*Proof.* Let  $\mathfrak{a}$  be an ideal of  $A$  and  $r(\mathfrak{a})$  be its radical ideal. Now, as  $A$  is Noetherian and  $r(\mathfrak{a})$  is an ideal of it we have  $r(\mathfrak{a}) = (a_1, a_2, \dots, a_t)$  for some  $a_1, a_2, \dots, a_t \in r(\mathfrak{a})$ . Next, for each  $a_i \exists n_i > 0$  such that  $a_i^{n_i} \in \mathfrak{a}$ . Take  $n = n_1 + n_2 + \dots + n_t$

Claim:  $r(\mathfrak{a})^n \subseteq \mathfrak{a}$

Clearly,  $r(\mathfrak{a})^n$  is generated by  $S := \left\{ a_1^{r_1} a_2^{r_2} \dots a_t^{r_t} \mid \sum_{i=1}^t r_i = n \right\}$

Now,  $\sum_{i=1}^t r_i = n = \sum_{i=1}^t n_i \Rightarrow \exists j \in \{1, 2, \dots, t\}$  such that  $r_j \geq n_j$

$$\Rightarrow a_1^{r_1} a_2^{r_2} \dots a_j^{r_j} \dots a_t^{r_t} \in \mathfrak{a}$$

$$\Rightarrow S \subseteq \mathfrak{a}$$

$$\Rightarrow r(\mathfrak{a})^n \subseteq \mathfrak{a} \quad \square$$

**Corollary 8.1.** *In a Noetherian ring the nilradical  $\mathfrak{R}$  is nilpotent*

*Proof.* Take  $\mathfrak{a} = 0$ . Then  $r(\mathfrak{a}) = \mathfrak{R}$  and thus  $\mathfrak{R}^n = 0$  for some  $n > 0$ . □

*We have now developed all the necessary tools to prove the ‘fundamental classification’ of Artinian rings. The theorem asserts that a ring is Artinian if and only if it is both*

Noetherian and of Krull dimension 0. The intuition behind this result is that Noetherian rings already exhibit strong control over ascending chains of ideals, while the additional assumption of dimension 0 ensures that prime ideals cannot be strictly contained in one another—forcing all prime ideals to be maximal. This restriction mimics the descending chain condition (dcc) found in Artinian rings, allowing us to conclude that the ring is Artinian. Conversely, if a ring is Artinian, it satisfies dcc on ideals, which somehow also prevents infinitely increasing complexity and ultimately forces it to be Noetherian. This characterization provides a deep structural insight into Artinian rings and highlights their close relationship with Noetherian rings.

**Theorem (Main).** *A ring is Artinian iff it is Noetherian and is of dimension 0*

*Proof.* ( $\Rightarrow$ ) By Lemma 5 every prime ideal is maximal. Thus every chain of prime ideals in  $A$  can have at most length 0 i.e., every chain of prime ideal is of length 0. Hence  $\dim A = 0$ . Next, as  $A$  is Artinian by Corollary 5.1 and Lemma 6 we have

$$\mathfrak{R} = \bigcap_{i=1}^n \mathfrak{m}_i$$

where  $\{\mathfrak{m}_i\}_{i=1}^n$  is the set of all maximal ideals of  $A$ . By Lemma 7  $\exists k > 0$  such that  $\mathfrak{R}^k = 0$

$$\begin{aligned} \Rightarrow \prod_{i=1}^n \mathfrak{m}_i^k &\subseteq \bigcap_{i=1}^n \mathfrak{m}_i^k \subseteq \left( \bigcap_{i=1}^n \mathfrak{m}_i \right)^k = 0 \\ \Rightarrow \prod_{i=1}^n \mathfrak{m}_i^k &= 0 \end{aligned}$$

i.e., the zero ideal is a product of maximal ideals (not necessarily distinct). Hence by Corollary 4.1 we conclude that  $A$  is Noetherian.

( $\Leftarrow$ ) Observe that since  $\dim A = 0$ , every chain of prime ideal is of length 0. Hence every prime ideal is maximal. Now, as 0 has a primary decomposition,  $A$  has only finitely many minimal prime ideals but here every prime ideal of  $A$  is minimal. Thus  $A$  has finitely many prime ideals say  $\{\mathfrak{m}_i\}_{i=1}^n$ . Hence

$$\mathfrak{R} = \bigcap_{i=1}^n \mathfrak{m}_i$$

By Corollary 8.1

$$\mathfrak{R}^k = 0 \Rightarrow \prod_{i=1}^n \mathfrak{m}_i^k = 0$$

Hence by Corollary 4.1  $A$  is Artinian □